

Online Appendix for “Log with zeros? Some problems and solutions”

Jiafeng Chen
Harvard Business School
jiafengchen@g.harvard.edu

Jonathan Roth
Brown University
jonathan_roth@brown.edu

November 17, 2023

Contents

A	Proofs of results in the main text	1
A.1	Proof of Proposition 1	1
A.2	Proof of Proposition 2	2
B	Extensions	4
B.1	Sensitivity to finite changes in scale	4
B.2	Extension to continuous treatments	5
B.3	Extension to OLS estimands and standard errors	6
C	Connection to structural equations models	9
C.1	Bellemare and Wichman (2020) and Thakral and Tô (2023)	10
C.2	Cohn et al. (2022)	11
C.3	Tobit models	12
D	Connection to two-part models	13
E	Details on Lee bounds using IV in Berkouwer and Dean (2022)	14
F	Appendix Tables and Figures	15

A Proofs of results in the main text

A.1 Proof of Proposition 1

Proposition 1. *Suppose that:*

1. *(The function m is continuous and increasing) $m : [0, \infty) \rightarrow \mathbb{R}$ is a continuous, weakly increasing function.*
2. *(The function m behaves like $\log$ for large values) $m(y)/\log(y) \rightarrow 1$ as $y \rightarrow \infty$.*
3. *(Treatment affects the extensive margin) $P(Y(1) = 0) \neq P(Y(0) = 0)$.*
4. *(Finite expectations) $E_{P_{Y(d)}}[|\log(Y(d))| \mid Y(d) > 0] < \infty$ for $d = 0, 1$.⁴⁶*

Then, for every $\theta^* \in (0, \infty)$, there exists an $a > 0$ such that $|\theta(a)| = \theta^*$. In particular, $\theta(a)$ is continuous with $\theta(a) \rightarrow 0$ as $a \rightarrow 0$ and $|\theta(a)| \rightarrow \infty$ as $a \rightarrow \infty$.

Proof. Note that $\theta(0) = E_P[m(0)] - E_P[m(0)] = 0$. Additionally, Proposition 4 below implies that $|\theta(a)| \rightarrow \infty$ as $a \rightarrow \infty$. To establish the proof, it thus suffices to show that $\theta(a)$ is continuous on $[0, \infty)$. The desired result is then immediate from the intermediate value theorem.

To establish continuity, fix some $a \in [0, \infty)$ and consider a sequence $a_n \rightarrow a$. Without loss of generality, assume $a_n < a + 1$ for all n . Let $m_{a_n}(y) = m(a_n y)$. Since m is continuous, $m_{a_n}(y) \rightarrow m_a(y)$ pointwise. We are done if we can apply the dominated convergence theorem to show that therefore $E[m_{a_n}(Y)] \rightarrow E[m_a(Y)]$.

Since $m(y)/\log(y) \rightarrow 1$ as $y \rightarrow \infty$, there exists $\bar{y}$ such that $m(y) < 2\log(y)$ for all $y \geq \bar{y}$. From the monotonicity of m , it follows that

$$\begin{aligned} m(0) \leq m(y) &\leq \mathbb{1}[y \leq \bar{y}]m(\bar{y}) + \mathbb{1}[y > \bar{y}]2\log(y) \\ &\leq \eta + 2 \cdot \mathbb{1}[y > \bar{y}]\log(y), \end{aligned} \tag{A.1}$$

where $\eta = |m(\bar{y})|$, and hence

$$\begin{aligned} m(0) \leq m_{a_n}(y) &\leq \eta + 2 \cdot \mathbb{1}[a_n y > \bar{y}]\log(a_n y) \\ &\leq \eta + 2 \cdot \mathbb{1}[y > 0] \cdot (|\log(a + 1)| + |\log(y)|) =: \bar{m}(y). \end{aligned}$$

for all n . Hence, we have that $|m_{a_n}(y)| \leq |m(0)| + \bar{m}(y)$ for all n , and the bounding function is integrable for $Y(d)$ for $d = 0, 1$ by the fourth assumption of the proposition. It follows from the dominated convergence theorem that $E_P[m_{a_n}(Y(d))] \rightarrow E_P[m_a(Y(d))]$ for $d = 0, 1$, and thus $\theta(a_n) \rightarrow \theta(a)$, as we wished to show. $\square$

⁴⁶This assumption simply ensures that $E_{P_{Y(d)}}[|m(aY(d))| \mid Y > 0]$ exists for all values of $a > 0$.

A.2 Proof of Proposition 2

Proposition 2 (A trilemma). *The following three properties cannot hold simultaneously:*

- (a) $\theta_g = E_P[g(Y(1), Y(0))]$ for a non-constant function $g : [0, \infty)^2 \rightarrow \mathbb{R}$ that is weakly increasing in its first argument.
- (b) The function g is scale-invariant.
- (c) θ_g is point-identified over $\mathcal{P}_+$.⁴⁷

Proof. To establish the proof of Proposition 2, we rely on Proposition 3, which shows that the only scale-invariant parameter of the form $E_P[g(Y(1), Y(0))]$ that is identified over distributions on the positive reals is the ATE in logs (up to an affine transformation).

Given Proposition 3, note that if $g : [0, \infty)^2 \rightarrow \mathbb{R}$ is increasing in y_1 , then it cannot be equal to $c \log(y_1/y_0) + d$ for $c > 0$ everywhere on $(0, \infty)^2$, since this would imply that $\lim_{y_1 \rightarrow 0} g(y_1, 1) = -\infty < g(0, 1)$. Proposition 2 is then immediate from Proposition 3, which shows that if properties (a) and (b) are satisfied, and θ_g is point-identified over $\mathcal{P}_{++} \subset \mathcal{P}_+$, then $g = c \log(y_1/y_0) + d$ on $(0, \infty)^2$. Thus, there does not exist such a g . $\square$

Proposition 3. *Let $\mathcal{P}_{++}$ denote the set of distributions over compact subsets of $(0, \infty)^2$. Suppose $g : (0, \infty)^2 \rightarrow \mathbb{R}$ is weakly increasing in y_1 and scale-invariant. Then θ_g is point-identified over $\mathcal{P}_{++}$ if and only if $g(y_1, y_0) = c \cdot (\log(y_1) - \log(y_0)) + d$, for constants $c \geq 0$ and $d \in \mathbb{R}$.*

Proof. We first show that point-identification over $\mathcal{P}_{++}$ implies that $g(\cdot, \cdot)$ must be additively separable. We do so by considering the points $\{y_0, y_0 + b\} \times \{y_1, y_1 + a\}$ on a rectangular grid. If $g(\cdot, \cdot)$ is not additively separable, then its expectation with respect to distributions supported on the rectangular grid depends on the correlation. Similar arguments appear in, e.g., Fan et al. (2017).

Formally, suppose that there exist positive values $y_1, y_0, a, b > 0$ such that

$$g(y_1, y_0) + g(y_1 + a, y_0 + b) \neq g(y_1 + a, y_0) + g(y_1, y_0 + b).$$

Now, consider the marginal distributions $P_{Y(d)}$ such that $P(Y(1) = y_1) = \frac{1}{2} = P(Y(1) = y_1 + a)$ and $P(Y(0) = y_0) = \frac{1}{2} = P(Y(0) = y_0 + b)$. Let P_1 and P_2 denote the joint distributions corresponding with these marginals and perfect positive and negative correlation of the potential outcomes, respectively. Then we have that

$$\begin{aligned} E_{P_1}(g(Y(1), Y(0))) &= \frac{1}{2} (g(y_1, y_0) + g(y_1 + a, y_0 + b)) \\ &\neq \frac{1}{2} (g(y_1 + a, y_0) + g(y_1, y_0 + b)) \end{aligned}$$

⁴⁷A minor technical complication arises from the fact that $E_P[g(Y(1), Y(0))]$ could be infinite for some P . For the purposes of our result, it suffices to trivially define θ_g to be identified in this case. Alternatively, the same result holds if part (c) is modified to impose only that θ_g is point-identified over all distributions in $\mathcal{P}_+$ with finite support, thus avoiding issues related to undefined expectations.

$$= E_{P_2}(g(Y(1), Y(0))),$$

and thus θ_g is not point-identified from the marginals at P_1 . Hence, if θ_g is identified over $\mathcal{P}_{++}$, then it must be that

$$g(y_1, y_0) + g(y_1 + a, y_0 + b) = g(y_1 + a, y_0) + g(y_1, y_0 + b) \text{ for all } y_1, y_0, a, b > 0,$$

and hence

$$g(y_1 + a, y_0) - g(y_1, y_0) = g(y_1 + a, y_0 + b) - g(y_1, y_0 + b) \text{ for all } y_1, y_0, a, b > 0.$$

It follows that we can write $g(y_1, y_0) = r(y_1) + q(\frac{1}{y_0})$, where $r(y_1) = g(y_1, 1) - g(1, 1)$ and $q(\frac{1}{y_0}) = g(1, y_0)$.

Second, we show that homogeneity of degree zero, combined with monotonicity, implies that g must be a difference in logarithms. Observe that since g is scale-invariant,

$$g(y_1, y_0) = g\left(\frac{y_1}{y_0}, \frac{y_0}{y_0}\right) = g\left(\frac{y_1}{y_0}, 1\right) =: h\left(\frac{y_1}{y_0}\right),$$

where h is an increasing function. We thus have that for any $a, b > 0$,

$$\begin{aligned} g(1, 1) &= h(1) = r(1) + q(1) \\ g(a, 1) &= h(a) = r(a) + q(1) \\ g\left(1, \frac{1}{b}\right) &= h(b) = r(1) + q(b) \\ g\left(a, \frac{1}{b}\right) &= h(ab) = r(a) + q(b) \end{aligned}$$

and hence $h(ab) = h(a) + h(b) - h(1)$. It follows that $\tilde{h}(x) = h(x) - h(1)$ is an increasing function such that $\tilde{h}(ab) = \tilde{h}(a) + \tilde{h}(b)$ for all $a, b \in \mathbb{R}$, i.e. an increasing function satisfying Cauchy's logarithmic function equation: $\phi(ab) = \phi(a) + \phi(b)$ for all positive reals a, b . Recall that if a function is increasing, then it has countably many discontinuity points, and thus is continuous somewhere. It is a well-known result in functional equations that the only solutions to Cauchy's logarithmic equation are of the form $\phi(t) = c \log(t)$, if we require that these solutions are continuous at some point; see [Aczél \(1966\)](#), Theorem 2 in Section 2.1.2.⁴⁸ Since we require monotonicity, the constant $c \geq 0$. Thus, $g(y_1, y_0) = h(y_1/y_0) = \tilde{h}(y_1/y_0) + \tilde{h}(1) = c \log(y_1) - c \log(y_0) + \tilde{h}(1)$. Letting $d = \tilde{h}(1)$ completes the proof of [Proposition 3](#). $\square$

⁴⁸Correspondingly, non-trivial solutions to Cauchy's logarithmic equations are highly ill-behaved.